Short | Box-Cox gamma sweep | Test ANN10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0342	0.0989	1.03	0.903	0.189	0.975	0.0714	0.988	1.01	0	18/18
1 ANN10	0.1	0.0321	0.142	1.26	1.01	0.17	0.99	0.0814	1.02	1.21	0	18/18
1 ANN10	0.2	0.0704	0.267	0.996	0.983	0.555	1	0.174	1.09	1.09	0	18/18
1 ANN10	0.3	0.0308	0.273	0.528	1.33	0.233	0.927	0.0761	2.59	5.57	0	18/18
1 ANN10	0.5	0.0607	0.52	0.763	0.94	0.507	0.808	0.0989	1.06	1	0	18/18
1 ANN10	1	0.0878	0.377	1.07	0.99	0.598	0.965	0.179	1	18.5	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Test ANN10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0542	0.129	0.937	0.943	0.179	1.28	0.137	0.555	0.805	0	18/18
1 ANN10	0.1	0.0137	0.114	0.555	3.71	0.208	0.91	0.0808	0.999	1.41	0	18/18
1 ANN10	0.2	0.036	0.151	0.694	1.11	0.223	1	0.121	0.665	0.931	0	18/18
1 ANN10	0.3	0.0183	0.134	0.358	0.973	0.167	0.784	0.046	0.657	2.01	0	18/18
1 ANN10	0.5	0.0351	0.274	0.584	0.499	0.299	0.499	0.082	0.787	0.977	0	18/18
1 ANN10	1	0.19	0.579	1.39	0.914	0.726	1.02	0.592	1	17.7	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Test P-only DeepONet10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
2 P-only DeepONet10	0.05	0.051	0.188	0.946	1.56	0.243	1.42	0.164	0.935	0.999	0	18/18
2 P-only DeepONet10	0.1	0.0578	0.114	0.951	0.994	0.165	0.993	0.123	0.995	1.19	0	18/18
2 P-only DeepONet10	0.2	0.0727	0.368	1	0.961	0.202	0.857	0.208	0.987	1.01	0	18/18
2 P-only DeepONet10	0.3	0.168	0.575	1	1	0.533	2.17	0.516	0.99	0.999	0	18/18
2 P-only DeepONet10	0.5	0.342	0.769	3	1.57	0.773	1.48	0.498	0.997	13.3	0	18/18
2 P-only DeepONet10	1	0.513	0.518	1.34	1.97	1.17	1.02	0.919	1.06	21.5	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Test P-only DeepONet10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
2 P-only DeepONet10	0.05	0.0521	0.216	1	0.78	0.236	0.871	0.0721	0.82	1	0	18/18
2 P-only DeepONet10	0.1	0.0217	0.0469	0.317	0.496	0.0973	0.108	0.0404	0.434	0.956	0	18/18
2 P-only DeepONet10	0.2	2.40e-03	0.0241	0.181	0.325	0.0392	0.0971	0.0192	0.637	0.765	0	18/18
2 P-only DeepONet10	0.3	6.50e-03	0.0497	0.776	0.524	0.238	0.335	0.0518	0.952	0.999	0	18/18
2 P-only DeepONet10	0.5	0.0781	0.482	1.31	1.15	0.628	0.995	0.247	1.03	1.48	0	18/18
2 P-only DeepONet10	1	0.666	0.639	17.8	21	0.909	5.89	0.467	1.42	811	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Test ANN10 + P-only DeepONet10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
3 ANN10 + P-only DeepONet10	0.05	0.0509	0.17	0.989	0.938	0.263	0.994	0.128	0.975	0.999	0	18/18
3 ANN10 + P-only DeepONet10	0.1	0.0373	0.0777	0.901	0.921	0.244	0.993	0.0941	0.995	1	0	18/18
3 ANN10 + P-only DeepONet10	0.2	0.0352	0.263	1	0.677	0.17	0.629	0.0924	0.975	0.998	0	18/18
3 ANN10 + P-only DeepONet10	0.3	0.0502	0.347	0.976	0.953	0.266	0.833	0.158	0.986	0.998	0	18/18
3 ANN10 + P-only DeepONet10	0.5	0.128	0.513	1.25	1.26	0.602	1.6	0.192	0.915	14.2	0	18/18
3 ANN10 + P-only DeepONet10	1	0.116	0.427	0.939	0.98	0.66	0.934	0.52	1.08	16.8	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Test ANN10 + P-only DeepONet10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
3 ANN10 + P-only DeepONet10	0.05	0.0843	0.17	2.36	0.722	0.154	0.835	0.0463	0.614	1	0	18/18
3 ANN10 + P-only DeepONet10	0.1	5.81e-03	0.0348	0.207	0.359	0.0715	0.101	0.0248	0.341	0.694	0	18/18
3 ANN10 + P-only DeepONet10	0.2	1.66e-03	0.02	0.11	0.225	0.0345	0.0917	0.0122	0.421	0.523	0	18/18
3 ANN10 + P-only DeepONet10	0.3	2.32e-03	0.0323	0.375	0.464	0.105	0.13	0.0411	0.861	0.999	0	18/18
3 ANN10 + P-only DeepONet10	0.5	0.0887	0.35	0.937	1.04	0.599	1	0.2	1.14	1.88	0	18/18
3 ANN10 + P-only DeepONet10	1	0.432	0.837	2.39	10.1	0.66	2.73	0.682	1.5	341	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Test T/P DeepONet10: common ICs [5, 7, 18, 39, 44, 45, 53, 57, 64, 68, 69, 76, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
4 T/P DeepONet10	0.05	0.291	0.413	1	1.16	0.707	1.07	0.356	2.27	1.84	0	18/18
4 T/P DeepONet10	0.1	0.0871	0.212	1.15	0.786	0.519	1.03	0.174	1.21	1	0	17/18
4 T/P DeepONet10	0.2	0.0278	0.115	0.881	1	0.425	1	0.351	0.996	1	0	18/18
4 T/P DeepONet10	0.3	0.404	1.29	0.964	13.6	0.929	1	0.587	1	1	0	18/18
4 T/P DeepONet10	0.5	0.244	0.646	2.1	1.17	0.728	1.02	0.458	11.4	4.69	0	16/18
4 T/P DeepONet10	1	0.433	1.37	1	3.23	0.906	0.658	0.725	1	55.8	0	17/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Test T/P DeepONet10: common ICs [5, 7, 18, 39, 44, 45, 53, 57, 64, 68, 69, 76, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
4 T/P DeepONet10	0.05	0.347	0.498	6.95	1.09	0.44	4.28	0.431	3.88	3.18	0	18/18
4 T/P DeepONet10	0.1	0.147	0.154	2.2	0.594	0.414	2.43	0.16	1.06	0.999	0	18/18
4 T/P DeepONet10	0.2	0.0194	0.188	0.624	1	0.623	0.43	0.188	1	1	0	18/18
4 T/P DeepONet10	0.3	7.50e-03	0.0612	0.22	0.287	0.231	0.137	0.124	0.908	0.703	0	18/18
4 T/P DeepONet10	0.5	0.239	1.48	1	1.05	0.929	1	1	1	1	0	18/18
4 T/P DeepONet10	1	0.363	0.945	6.73	2.1	0.842	1.21	0.569	4.76	15.8	0	16/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Test ANN10 + T/P DeepONet10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
5 ANN10 + T/P DeepONet10	0.05	0.0747	0.616	1	0.962	0.563	1.03	0.18	1.5	2.4	0	18/18
5 ANN10 + T/P DeepONet10	0.1	0.0491	0.0957	0.962	0.82	0.174	0.97	0.066	0.921	0.993	0	18/18
5 ANN10 + T/P DeepONet10	0.2	7.87e-03	0.0701	0.265	0.595	0.333	0.238	0.0622	0.996	0.999	0	18/18
5 ANN10 + T/P DeepONet10	0.3	0.0117	0.0733	0.286	0.646	0.365	0.196	0.103	1	1	0	18/18
5 ANN10 + T/P DeepONet10	0.5	0.16	0.499	1.42	3.36	0.699	1.4	0.313	6.87	5.37	0	18/18
5 ANN10 + T/P DeepONet10	1	0.242	0.721	2.28	2.68	0.749	0.599	0.285	1.08	16.6	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Test ANN10 + T/P DeepONet10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
5 ANN10 + T/P DeepONet10	0.05	0.213	0.167	2.17	0.894	0.294	5.8	0.275	1.34	2.35	0	18/18
5 ANN10 + T/P DeepONet10	0.1	0.185	0.135	6.18	0.731	0.298	4.5	0.14	1.29	0.991	0	18/18
5 ANN10 + T/P DeepONet10	0.2	9.15e-03	0.0452	0.487	0.699	0.459	0.194	0.0626	0.994	0.998	0	18/18
5 ANN10 + T/P DeepONet10	0.3	7.93e-03	0.0398	0.152	0.376	0.299	0.0964	0.0212	0.914	0.941	0	18/18
5 ANN10 + T/P DeepONet10	0.5	0.0562	0.991	1	1	0.568	0.677	0.335	1	1	0	18/18
5 ANN10 + T/P DeepONet10	1	0.388	0.868	8.01	5.99	0.925	1.7	0.28	3.69	67.5	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Test T/P/ϕ DeepONet10: common ICs [4, 5, 32, 39, 44, 45, 57, 64, 68, 74, 76, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
6 T/P/ϕ DeepONet10	0.05	0.15	0.215	1.01	0.996	0.438	1.38	0.191	1.41	0.999	0	18/18
6 T/P/ϕ DeepONet10	0.1	0.108	0.152	0.962	1	0.564	0.986	0.133	1.25	1.46	0	17/18
6 T/P/ϕ DeepONet10	0.2	0.062	0.45	0.528	0.842	0.517	0.866	0.0767	0.995	1	0	18/18
6 T/P/ϕ DeepONet10	0.3	0.0965	0.896	1.18	0.885	0.723	0.981	0.108	0.992	0.951	0	18/18
6 T/P/ϕ DeepONet10	0.5	0.213	0.601	2.18	3.81	0.763	2.38	0.282	1.56	1.5	0	18/18
6 T/P/ϕ DeepONet10	1	0.274	0.905	1.6	2.52	0.847	0.669	0.32	1.09	6.39	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Test T/P/ϕ DeepONet10: common ICs [4, 5, 32, 39, 44, 45, 57, 64, 68, 74, 76, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
6 T/P/ϕ DeepONet10	0.05	0.38	0.403	5.59	1.03	0.384	5.92	0.379	1.41	1.85	0	18/18
6 T/P/ϕ DeepONet10	0.1	0.504	0.287	7.37	0.689	0.404	5.34	0.364	1.62	2.66	0	18/18
6 T/P/ϕ DeepONet10	0.2	0.0642	0.481	0.88	0.532	0.62	0.402	0.0802	1	1	0	18/18
6 T/P/ϕ DeepONet10	0.3	0.217	1.07	1	1	0.696	0.906	0.361	0.999	1	0	18/18
6 T/P/ϕ DeepONet10	0.5	0.291	0.854	0.954	1.13	0.768	1.12	0.39	0.979	2	0	18/18
6 T/P/ϕ DeepONet10	1	0.285	0.96	4.98	4.06	0.73	1.21	0.281	1.18	19.8	0	13/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Test ANN10 + T/P/ϕ DeepONet10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0836	0.331	0.998	0.995	0.489	1.72	0.382	0.998	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0633	0.0986	1	0.584	0.195	0.998	0.192	0.998	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0124	0.124	0.476	0.512	0.378	0.288	0.0502	0.997	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0468	0.334	1	0.783	0.401	0.943	0.0996	0.994	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.122	0.618	3.29	1.83	0.622	1.11	0.208	1.39	0.997	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	1	0.206	0.646	2.83	1.65	0.759	0.441	0.219	1.35	13.8	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Test ANN10 + T/P/ϕ DeepONet10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.335	0.45	2.98	0.931	0.329	8.17	0.569	1.01	1.66	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.14	0.155	1.69	0.696	0.0989	2.96	0.194	0.96	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0344	0.28	1.01	0.479	0.487	0.46	0.0813	0.998	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.136	0.558	1.02	1.98	0.644	0.689	0.264	1	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.139	0.205	0.999	1.19	0.305	0.875	0.385	1	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	1	0.337	0.804	7.55	5.53	0.881	3.68	0.36	4.65	56.3	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.